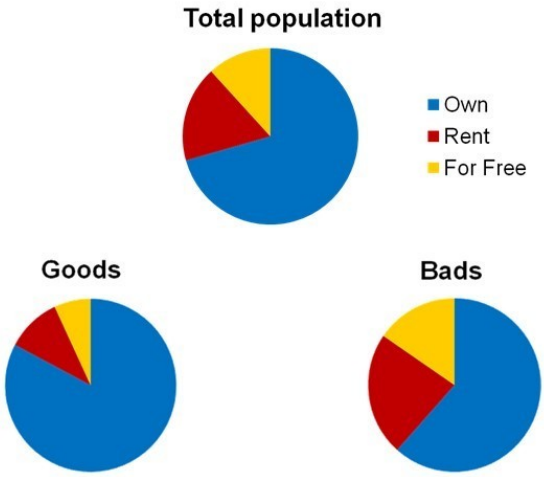
Chapter 2: Preprocessing

Multiple choice questions

Question 1

Level: easy

The below figure provides information on the housing variable in a credit risk dataset, with pie charts for the full customer base and separately for the good (non-defaulted) and bad (defaulted) observations. Which statement about the below figure is **TRUE**?



a) The housing variable is predictive for the target variable, since the distributions of the goods and bads are different.
b) Default risk for a customer living for free is lower than for a customer who is an owner.
c) A good customer is less likely to be an owner than a renter.
d) A bad customer is more likely to be an owner than a renter and therefore an owner is more likely to be bad.

Question 2

Level: difficult

Which of the following statements is **TRUE**? Give answer D if all statements are false.

a) When the class distribution is heavily skewed, you will automatically obtain more reliable or robust weights-of-evidence values if more observations are used for calculating weights-of-evidence values.
b) When applying weights-of-evidence, you can expect to lose predictive power, no matter which classification method is applied.
c) If we can have at most five variables in our final model, then selecting the five variables with the highest information value will lead to the best predictive performance of the resulting model.
d) All the above statements are false.

Open questions

Question 1

Why do categorical variables need to be transformed into continuous variables? Is that always necessary? Discuss using an example.

Question 2

Why do continuous variables need to be transformed into categorical variables? Is that always necessary? Discuss using an example.

Question 3

Explain why and when the log transformation is used?

Question 4

What is in your opinion the best approach to handle missing values?

Question 5

Why is it important to detect and treat outliers in preprocessing a data set? Is it always necessary or recommendable to do that? Discuss using an example.